EXPERIMENT 4

Azure Databricks for Data Preprocessing and Machine Learning

Aim

To use Azure Databricks for data preprocessing and machine learning by building a complete PySpark MLlib pipeline that cleans, transforms, and trains classification models on a student performance dataset with MLflow experiment tracking.

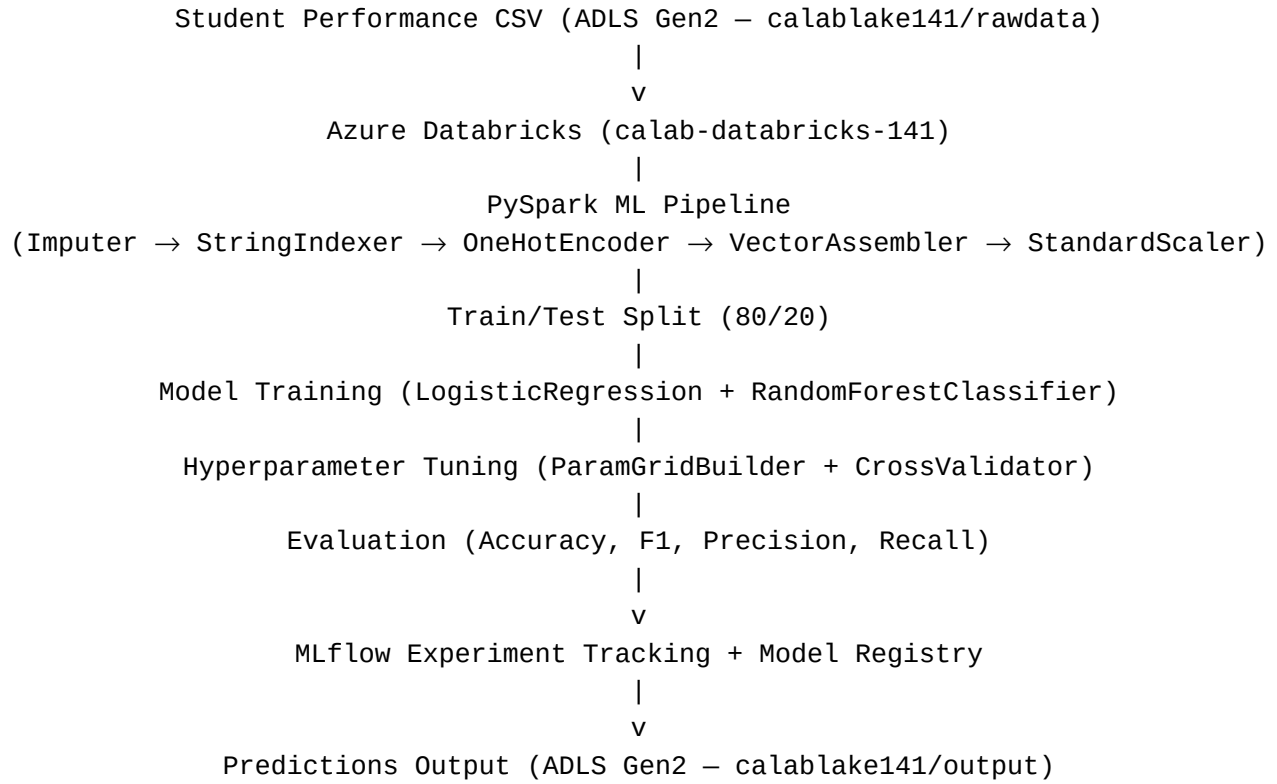
Theory

Azure Databricks is an Apache Spark-based analytics platform optimized for Azure. It provides a collaborative notebook environment for big data processing and machine learning. Databricks integrates with Azure Data Lake Storage Gen2 for scalable data access and with MLflow for experiment tracking and model management.

PySpark MLlib is the machine learning library for Apache Spark, providing scalable implementations of common ML algorithms. The ML pipeline API (pyspark.ml) enables building reproducible feature engineering and model training workflows using Transformers, Estimators, and Pipelines.

In this experiment, a student performance dataset (Abhi, Kshitiz, Aakriti, Rohit, Sandhya, Riya) is loaded from ADLS Gen2 into a Databricks notebook. The data is profiled for missing values and class distribution. A preprocessing pipeline is built using Imputer (median strategy), StringIndexer, OneHotEncoder, VectorAssembler, and StandardScaler. Logistic Regression and Random Forest models are trained and compared. Hyperparameter tuning is performed using ParamGridBuilder with 3-fold CrossValidator. All experiments, parameters, metrics, and models are logged to MLflow.

Flow Diagram



Procedure

1. Open Azure Databricks workspace calab-databricks-141 from the Azure Portal.
2. Create a single-node compute cluster (ml-cluster-141) with Runtime 14.3 LTS and Standard_DS3_v2.
3. Upload student_performance.csv to DBFS or mount ADLS Gen2 container calablake141/rawdata.
4. Import Exp04_Databricks_Preprocessing_ML.py as a Databricks notebook.
5. Run Cell 1: Load CSV data, print schema, display dataset (50 rows, 13 columns).
6. Run Cell 2: Data profiling — summary statistics, null counts per column, FinalGrade distribution.
7. Run Cell 3: Build preprocessing pipeline — Imputer (median), StringIndexer, OneHotEncoder, VectorAssembler, StandardScaler.
8. Run Cell 4: Split data 80/20, fit preprocessing pipeline on training set.
9. Run Cell 5: Train Logistic Regression and Random Forest classifiers.
10. Run Cell 6: Evaluate models — compare Accuracy, F1, Precision, Recall metrics.
11. Run Cell 7: Hyperparameter tuning with ParamGridBuilder (numTrees: 50/100/200, maxDepth: 3/5/10) and 3-fold CrossValidator.
12. Run Cell 8: Log best model parameters, metrics, and artifact to MLflow.
13. Run Cells 9-10: Display feature importances, confusion matrix, save predictions to ADLS output.

Expected Output

<GitHub Link Here>

Result

Thus, data preprocessing and machine learning were successfully performed using Azure Databricks. The student performance dataset was cleaned using PySpark Imputer and encoded using StringIndexer and OneHotEncoder. Logistic Regression and Random Forest classification models were trained and compared. Hyperparameter tuning via CrossValidator improved model accuracy. All experiment runs, parameters, metrics, and models were tracked using MLflow. The predictions were saved to ADLS Gen2 in Parquet format. This experiment demonstrates a scalable, cloud-native ML workflow widely used in enterprise data science on Azure.